

Gabrielle Pang

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EDUCATION

NANYANG POLYTECHNIC

2023 - 2026

Diploma in Robotics and Engineering, School of Engineering

Singapore

HOLY INNOCENTS' HIGH SCHOOL

2018 - 2022

GCE O-Level

Singapore

TECHNICAL PROJECTS

TELEGRAM AI AGENT CHATBOT

2026

Personal Project | AI Agent, Chatbot Automation & API Integration

Singapore

- Developed a Telegram-based AI agent chatbot connected to a Small Language Model through API integration
- Built chatbot workflows to record tasks, notes and reminders through natural language conversations
- Integrated API key-based communication between Telegram and the model, refining responses for usability and task-tracking accuracy

CUSTOM ORDERING APP FOR SMALL BUSINESS

2026

Personal Project | Web App & Business Automation

Singapore

- Built a custom ordering web application for a small business, allowing customers to browse products, select quantities and submit orders digitally
- Developed responsive desktop and mobile interfaces with product, stock and order-management features to reduce manual tracking
- Tested real ordering scenarios and refined the app based on usability issues, layout feedback and business needs

IOT PEOPLE COUNTER - Smart IR Detection System

2025

Final Year Project | IoT, Sensor Systems & Cloud Integration

Singapore

- Engineered directional IR detection logic to classify and count individuals in real time while minimising false positives
- Integrated a multi-node sensor network with a centralised WiFi hub and connected IoT data to AWS for cloud-based transmission, storage and monitoring
- Validated outputs against real-world test cases and enabled live monitoring via web/mobile dashboard to improve reliability

ROBOT BARISTA ARM AUTOMATION - OITP Overseas Internship

2025

Team Project | Robotic Systems & Automation

Korea

- Applied GPIO-triggered control logic between Raspberry Pi and robotic arm, enabling precise and repeatable automated motion sequences
- Built RoboDK simulations to validate and annotate motion paths before physical deployment, improving positional accuracy and reducing trial errors
- Programmed and refined robotic drink-making sequences through repeated testing, visual inspection and troubleshooting

LEADERSHIP & ACTIVITIES

GEO COUNCIL - NANYANG POLYTECHNIC

2025

Member

Singapore

- Designed and produced visual content for student events; managed feedback cycles to meet quality and consistency standards

FLOORBALL TEAM - HOLY INNOCENTS' HIGH SCHOOL

2018 - 2022

CCA Team Member

Singapore

- Represented the school in B and C Division; qualified for B Division Round 2 in 2022, the first time in school history

AWARDS / ACHIEVEMENTS

Smart Parking System selected for NYP Open House showcase (2024) • Floorball B Division qualified for Round 2, first in school history (2022) • ActiveSG SFA Floorball C Division Plate & Bowl - 3rd Place (2018)

SKILLS & INTERESTS

AI & Automation: AI Agents, Telegram Bot Development, SLM/LLM API Integration, Prompt Engineering, Task Automation, Chatbot Workflows

AI & Vision: Image Processing, OpenCV, Feature Extraction, Visual Inspection, Pattern Recognition, Computer Vision, Annotation Quality Control

Cloud & IoT: AWS, IoT Data Transmission, Cloud Integration, Sensor Networks, ESP8266, Raspberry Pi, WiFi Hub Integration

Technical: GPIO, Arduino, RoboDK, Firmware Configuration, IoT Systems, Soldering, CAD Modelling

Software: API Integration, Web App Development, Responsive UI Design, Order Management Systems, GitHub, Frontend Development

Soft Skills & Languages: Attention to Detail, Analytical Thinking, Quality Control, Team Collaboration, Adaptability; English, Mandarin